

Lab 11 — The Video Agent Team: Script to YouTube

Multi AI Agents Workflow for Content Creation · Day 2 · Claude Code (agent team) → Codex (render, upload)

Course: Multi AI Agents Workflow for Content Creation (TGS-2023036153)

Day 2 · Topic 3 · about 40 minutes · slides 105–110

Surface: Claude Code (agent team) → Codex (render, upload)

Features: agent team for the script · approval of script and video · ffmpeg render · private YouTube upload

The story so far

Pre-retirees say they would rather watch than read. Rachel wants the first 'Money in Plain English' video: 60 seconds on the CPF retirement sums, uploaded privately until she has seen it.

Your goal

Video reaches the people who never read posts. A team writes and checks the script; you approve it; Codex makes the video and uploads it — privately, until you decide to make it public.

You'll build

content/video/ep01/: scenes.json, ep01.mp4, ep01.md, and a private YouTube upload (or dry run)

What is in this folder

- assets/youtube-brief.md
- assets/video-spec.md
- assets/sample-explainer.mp4
- solution/ — reference files from the verified build
- prompts.md / prompts.pdf — every prompt, ready to paste
- evidence/checklist.md — what to capture as proof

Step by step

1. **Script it as a team** — Paste Prompt A in Claude Code.
2. **Approve the script** — Read scenes.json and ep01.md, then approve yt-ep01-script in your own terminal.
3. **Render it** — In Codex, paste Prompt B. Watch the result.
4. **Approve the video** — Run submit.mjs on ep01.md, watch ep01.mp4 to the end, then approve yt-ep01.
5. **Upload** — Paste Prompt C. It goes up as private.

6. **Disclose** — In YouTube Studio, set the altered or synthetic content answer correctly before you make it public.

The prompts

PROMPT A — Claude Code: the script team

Create an agent team for YouTube episode 1 from data/youtube-brief.md. Spawn three teammates: researcher (agent type researcher), a video-scriptwriter and compliance-reviewer (agent type compliance-reviewer). The scriptwriter writes content/video/ep01/scenes.json (5-7 scenes: seconds, title, text) and ep01.md (id yt-ep01, channel youtube, title, tags, video content/video/ep01/ep01.mp4; the body is the description with chapters and the disclosure). Submit the script as yt-ep01-script once the reviewer passes it. Publish nothing.

PROMPT B — Codex: render

Following data/video-spec.md, write scripts/render-video.py and run it on the approved content/video/ep01/scenes.json to make content/video/ep01/ep01.mp4. Use ffmpeg and Pillow. Check the result with ffprobe: 1920x1080, under 90 seconds, every scene present. Do not change the approved words.

PROMPT C — Codex: upload

Run node scripts/publish.mjs yt-ep01 as a dry run and show me the request. If I say "go", run it with --live: it uploads the video as PRIVATE. Give me the video id and the YouTube Studio link.

Check your work

- ☐ The reviewer passed the script; you approved yt-ep01-script.
- ☐ ep01.mp4 is 1920x1080, under 90 seconds, every scene present.
- ☐ Every figure on screen matches facts-2026.md.
- ☐ The description has chapters, UTM link, disclaimer and AI disclosure.
- ☐ You approved yt-ep01 after watching it to the end.
- ☐ The upload is private (or a dry run printed the request).

If it goes wrong

- **ffmpeg not found** — Install it (brew install ffmpeg / winget install ffmpeg) or use sample-explainer.mp4 to test the upload.

- **Upload forbidden** — Add yourself as a test user on the OAuth consent screen, then refresh the token.

Stretch

- Add a 30-second vertical Short from the same scenes.

Why it matters: Approve twice: once for the words, once for the finished video. Rendering can break what the script got right.

Next

Lab 12 — Hermes Agent Bots on a Kanban Board. Keep what you built — the next lab starts from it.

Safety

Use only this lab's synthetic data and your own accounts. Never paste a real API key, password, token or client data into a prompt or a file — tokens live only in .env. Every email address in the data must resolve to your own inbox. Nothing is published or sent without a person approving that exact version.